Newton's divide and difference form

Limitation of Langrage: can't add new terms.

Solution

Newton's basis

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$
$$+ f[x_0, x_1, x_2, x_3](x - x_0)(x - x_1)(x - x_2) + \cdots$$
$$+ f[x_0, x_1 \ldots x_n](x - x_0)(x - x_1) \cdots (x - x_{n-1})$$

$$P_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

	x	f(x)
x_0	0	10
x_1	1	20
x_2	2	30
	5	?

$$P_2(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1)$$

$x_0 = 0$, $f[x_0] = 10$

$x_1 = 1$, $f[x_1] = 20$

$x_2 = 2$, $f[x_2] = 30$

$$f[x_0, x_1] = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = 10$$

$$f[x_1, x_2] = \frac{f[x_2] - f[x_1]}{x_2 - x_1} = 10$$

$$f[x_0, x_1, x_2] = \frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0} = 0$$

$$P_2(x) = 10 + 10(x - 0) + 0(x - 0)(x - 1) = 10 + 10x$$

[Polynomial expression of newton's method]

Now,

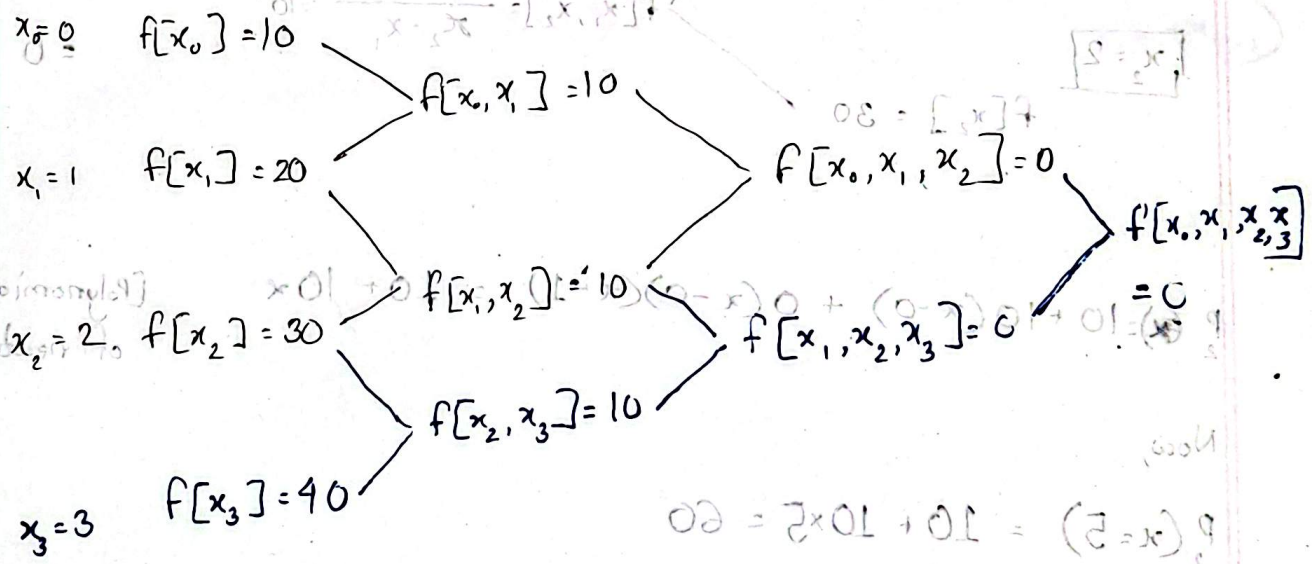
$$P_2(x = 5) = 10 + 10 \times 5 = 60$$

b =

$$\begin{bmatrix} y_0 = f[x_0] & f[x_0, x_1] & f[x_0, x_1, x_2] & f[x_0, x_1, x_2, x_3] \\ y_1 = f[x_1] & f[x_1, x_2] & f[x_1, x_2, x_3] & 0 \\ y_2 = f[x_2] & f[x_2, x_3] & 0 & 0 \\ y_3 = f[x_3] & 0 & 0 & 0 \end{bmatrix}$$

x	f(x)
x_0 0	10
x_1 1	20
x_2 2	30
x_3 3	40

$$P_3(x) = \underbrace{f[x_0]}_{a_0} + \underbrace{f[x_0, x_1]}_{a_1}(x - x_0) + \underbrace{f[x_0, x_1, x_2]}_{a_2}(x - x_0)(x - x_1) + \underbrace{f[x_0, x_1, x_2, x_3]}_{a_3}(x - x_0)(x - x_1)(x - x_2)$$



* We can reuse values if new terms are added

Sunday 5.7.2026

L-7 : Derivation

Newton's divide and diff formula:

$$P_n(x) = f[x_0] + f[x_0, x_1](x-x_0) + f[x_0, x_1, x_2](x-x_0)(x-x_1) + \dots$$
$$f[x_0, x_1, \dots, x_n](x-x_0)(x-x_1)\dots(x-x_n)$$

$$P_n(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots$$
$$= a_0 + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2)$$

Newton's polynomial expression

$f(x) =$ ___ + ___ + ___ + ___ + ...

$P_n(x) =$ ___ + ___ + ___ + ___ +

$$f(x) = P_n(x) + g_{n+1}$$

$$\Rightarrow g_{n+1} = f(x) - P_n(x)$$

$$\Rightarrow a_{n+1} = \frac{f(x) - P_n(x)}{(x-x_0)(x-x_1)\dots(x-x_n)}$$

coefficient of Newton's formula

$$g_{n+1} = a_{n+1}(x-x_0)(x-x_1)(x-x_2)\dots(x-x_n)$$

$$g_n = \prod_{j=0}^{k-1}(x-x_j)$$

Newton's basis

$$n_0 = 1$$
$$n_1 = (x-x_0)$$
$$n_2 = (x-x_0)(x-x_1)$$
$$n_3 = (x-x_0)(x-x_1)(x-x_2)$$

Newton's basis

$$P_n(x) = \sum_{n=0}^{K-1} q_k n_k$$

Polynomial n degree

$$= q_0 n_0 + q_1 n_1 + q_2 n_2 + \dots + q_n n_n$$

Newton's Formula

$$= q_0 1 + q_1 (x-x_0) + q_2 (x-x_0)(x-x_1) + \dots + q_n (x-x_0)(x-x_1)\dots(x-x_{n-1}) \quad \text{---(i)}$$

If $x = x_0$ in (i),

$$P_n(x) \Rightarrow c_0 = f[x_0]$$

If $x = x_1$ in (ii)

$$P_n(x_1) = c_0 + c_1(x_1 - x_0) + 0$$

$$\Rightarrow f[x_1] = f[x_0] + c_1(x_1 - x_0)$$

$$\Rightarrow c_1 = \frac{f[x_1] - f[x_0]}{x_1 - x_0} = f[x_1, x_0]$$

If $x = x_2$ in (i),

$$P_n(x_2) = c_0 + c_1(x_2 - x_0) + c_2(x_2 - x_0)(x_2 - x_1) + \dots = f[x_2]$$

$$\Rightarrow c_2 = \frac{f[x_2] - c_0 - c_1(x_2 - x_0)}{(x_2 - x_0)(x_2 - x_1)}$$

$$= \frac{f[x_2] - f[x_1] - \frac{f[x_1] - f[x_0]}{(x_1 - x_0)}(x_2 - x_0)}{(x_2 - x_0)(x_2 - x_1)}$$

$$\frac{f[x_1, x_2] - f[x_0, x_1]}{x_2 - x_0}$$

$$= f[x_0, x_1, x_2]$$

$$P_n(x) = f[x_0] + f[x_0, x_1](x - x_0) + f[x_0, x_1, x_2](x - x_0)(x - x_1) + \ldots\ldots$$

$n = 2$